A

GapMind prediction (*in silico*)

Pathway predicted present (+)

Pathway predicted absent (−)

Experimental outcome (*in vivo*)

Growth observed (+)

No growth (−)

a TRUE POSITIVE
concordant

BIOLOGICAL STATE

Genes present and functional
Pathway correctly annotated as present
Experiment accurately measures growth

INTERPRETATION

Prediction and phenotype agree:
gene-to-function mapping is supported.

High-quality training sample

b FALSE NEGATIVE
discordant

POSSIBLE CAUSES

ANNOTATION	homolog not detected
BIOLOGY	alternative pathway present
BIOLOGY	promiscuous enzyme activity
MEASUREMENT	false-positive growth call
MEDIA	growth on base media (carryover)

Growth via unknown mechanism

c FALSE POSITIVE
discordant

POSSIBLE CAUSES

REGULATION	pathway repressed under condition
MEDIA	insufficient incubation time
MEDIA	missing cofactors / inducers
ANNOTATION	false-positive homology call
BIOLOGY	pseudogene or non-functional allele

Genes present but not functional

d TRUE NEGATIVE
concordant

BIOLOGICAL STATE

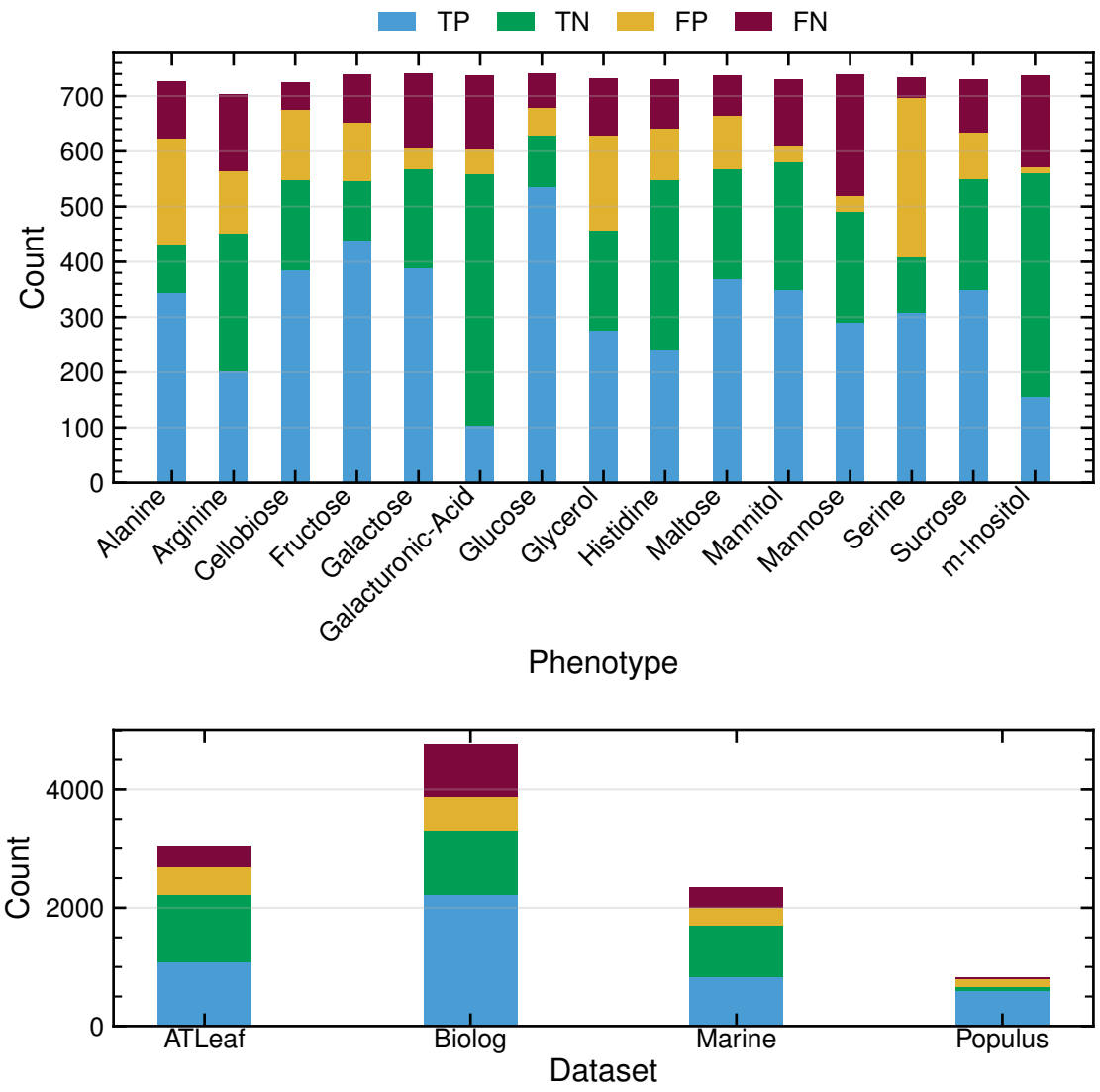
Genes absent from genome
Pathway correctly annotated as absent
No alternative pathway or leaky activity
Experiment accurately measures no-growth

INTERPRETATION

Absence of capability confirmed by both
genome content and phenotype.

High-quality training sample

B



C

